# DINESH NATARAJAN

**Aerospace Engineer | Engineering Task Author and Grader Reviewer for AI Training | FEA, CFD, CAD**

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## PROFILE

MSc aerospace engineer who builds and reviews terminal-based engineering tasks for AI training and evaluation: the problem statement, geometry and mesh, the reference solution, the automated grader and the containerised environment. Six Hardware/CAD tasks accepted on Handshake's Project Insight (Terminal-Bench 3 format) in October 2026, each with a separate verifier image, held-back cases and exact geometric grading; the last two were not solved in any of five trials by the benchmark's frontier model. Earlier, authored CFD and structural reinforcement-learning tasks at Alignerr (Labelbox), calibrated against repeated agent runs, and now reviews aerospace engineering solution sets at Innodata. Reviews for units, boundary conditions, mesh and solver convergence, reference frames and reproducibility, and for ways a submission can score without doing the engineering. Works in Python, Docker, Git and Linux with ANSYS, Abaqus, HyperMesh, OpenFOAM, CadQuery/OpenCascade and MATLAB.

## CORE SKILLS

- **Task and grader review:** problem statements, reference solutions, automated graders (stresses, deflections, mid-surface deviation, aerodynamic coefficients, safety margins), hidden evaluation cases, difficulty calibration against agent runs, bypass and reward-hacking review, reproducibility (pinned dependencies, deterministic sampling, Docker images), fail-closed numerics
- **Simulation validation:** units, equilibrium and load paths, boundary conditions, mesh-independence checks, solver health and convergence gates, reference frames, physically invalid results, time-out and runtime budgets
- **Domains:** finite-element analysis (static, modal, contact), CFD and aerodynamics (OpenFOAM, vortex lattice, airfoil analysis), structural dynamics and damping, flight dynamics (state-space), fatigue and damage tolerance, B-rep geometry and GD&T (ASME Y14.5)
- **Tools:** ANSYS Mechanical, Abaqus/Standard, Altair HyperMesh, OpenFOAM, XFLR5 (XFoil), AeroSandbox, OpenSeesPy, CadQuery and OCP (OpenCascade), PTC Creo, CATIA V5, MATLAB/Simulink
- **Software practice:** Python (NumPy, SciPy, pandas, pytest), Bash, Linux (Ubuntu), Docker, Git and GitHub pull requests, CI/CD gates, Harbor task format, AI coding agents (Claude Code, Codex)
- **Leadership and feedback:** led a 21-member operations team; led a three-person MSc design team; written, evidence-based review notes on every item rated

## PROFESSIONAL EXPERIENCE

### Task Author, Hardware/CAD (Contract)
**Sep 2026 to Oct 2026**

*Handshake AI, Project Insight | Remote*

- **Six accepted tasks:** authored six terminal-based CAD and structures tasks for a frontier-model benchmark; every one passed the automated review chain (static checks, artificiality, oracle and no-op validation, implementation rubric, deep review, adversarial verifier audit, Pass@2 and Pass@5 trials).
- **Task packages:** an instruction in the voice of a stress or manufacturing office, a disclosed engineering standard, a Harbor/Docker agent environment, and a separate verifier image holding the generating geometry and held-back cases pinned by SHA-256, with pytest graders, CTRF reports and a binary reward written only by the verifier.
- **Exact graders:** two-way B-rep distance at 1.0 mm sampling with a Lipschitz bound on the unsampled deviation, coverage, free-edge checks after sewing, thickness within 2 percent, and retained-solid count, volume and surface checks; non-finite values fail closed.
- **Bypass review and hardening:** the tool under test runs as a fresh unprivileged user under Landlock and seccomp with resource limits, verifier trees are root-only and the reward directory is sealed; reproduced CI hosts locally with seccomp profiles (no Landlock, restricted user namespaces) and fixed each finding in one commit.
- **Verifier batteries:** on one task, 54 cases: alternative valid constructions pass (split faces, strips, sewn shells, single face with wire holes, 0.03 mm shifts) while naive methods fail by 10 to 30 times the tolerance; band-edge cases (0.06 mm against a 0.05 mm band) and planted NaNs fail.
- **Tasks:** a GFEM mid-surface idealisation tool graded on five hidden machined parts and a predictive shim-release tool graded on five hidden assemblies (each 0 of 5 trials solved); an LPBF hydraulic manifold, a hob-true spur-gear re-ratio, a bump-formed sheet-metal transition and an as-machined closed-angle frame.

### Reinforcement Learning Task Author, Engineering Simulation (Contract)
**Jun 2026 to Sep 2026**

*Alignerr (Labelbox) | Remote*

- **Task specifications:** engineering design problems with defined inputs, constraints, operating conditions, output metrics and numeric acceptance criteria, framed around the required function so different design approaches can pass.

- **Reference and graders:** a verified reference solution per task scoring 1.000 against its own grader; deterministic Python graders that reject poor meshes, diverged solves and physically invalid results and score the worst case across operating conditions, including hidden evaluation cases.
- **Difficulty calibration:** retuned a wing-planform task from 0.90 to 0.62 against repeated frontier-agent runs; closed a geometry loophole; removed a ready-to-call model that let agents optimise straight to the reference; replaced an additive rubric floor that made the difficulty gate unreachable with multiplicative feasibility gating.
- **Agent run review:** diagnosed failures from transcripts (a read-only output file that blocked 5 of 5 runs; time-outs from long Monte Carlo sweeps) and reworked instructions so runs finish inside the limit (a 71-minute time-out became a 49-minute completion).
- **Portfolio and physics:** wing planform design by vortex lattice behind an OpenFOAM solver-health gate; backward-facing-step reattachment control in OpenFOAM, rebuilt after tracing an agent plateau to a contradictory spec key; OpenSeesPy seismic retrofits, where Rayleigh damping that over-damped under uniform ground excitation was replaced by explicit dashpots; all delivered through Docker, GitHub pull requests and CI/CD.

**Aerospace Engineering Expert, AI Training (Contract)** **Sep 2026 to Present**

*Innodata Inc., via Upwork | Remote*

- Review and rate aerospace engineering question, solution and answer sets used to train AI models, including problems built on engineering drawings; check each worked solution for technical accuracy and completeness and write a justification for every rating.

**Procurement and Inventory Operations Supervisor** **May 2023 to Sep 2025**

*Tops | Manchester, United Kingdom*

- Led a 21-member operations team running procurement, stock replenishment and inventory control; set daily priorities and tracked delivery across functions.

**Product Design Engineer (Intern)** **Oct 2020 to Apr 2021**

*Black Space Research and Development | Bangarpet, Karnataka, India*

- Built a shell-element FE model of a truck rear underrun protection device in Altair HyperMesh and solved three regulatory load cases with contact in Abaqus/Standard; identified the governing case (234.6 MPa against 230 MPa yield at the mounting flange) and recommended stronger material grades.

## ENGINEERING PROJECTS

**Aircraft Engine Bracket: Topology Optimisation and Fatigue Substantiation** **2023, extended 2026**

*MSc dissertation, University of Salford; independent extension | PTC Creo Parametric, ANSYS Mechanical, NASGRO*

- Two bracket concepts for the GE bracket challenge loads in SAE 4130; three static load cases plus modal analysis; mass cut 22 percent by topology optimisation at a factor of safety of 1.73; a mesh-independence check moved peak stress by under 1 percent; results cross-checked against hand calculations.
- 22-page substantiation: S-N, Goodman and Miner safe-life, NASGRO-equation crack growth and MIL-HDBK-1530 inspection intervals; GD&T and tolerance stack-up (worst case ±0.35 mm, RSS ±0.206 mm) in CATIA V5.

**Twin-Boom Surveillance UAV: Conceptual Design to 3D-Printed Model** **2022**

*MSc group project, University of Salford*

- Led the airfoil analysis, the design integration and the final report in a three-person team that sized a 2.2 m, 2.8 kg pusher UAV (constraint diagram, sizing matrix, carpet plot, CG, trim and stability checks); led the 3D printing and control-surface integration at the Salford Makerspace.

**Airfoil, Flight-Dynamics and Surrogate Studies** **2023 to 2025**

*Independent projects | XFLR5, OpenFOAM, MATLAB/Simulink, Python*

- Natural-laminar-flow comparison (NACA 63-215 against NACA 2412: peak L/D 67.3 against 46.8); Cessna 172 longitudinal state-space model meeting MIL-F-8785C short-period criteria with 6 of 6 V&V checks passed; neural-network surrogate for aerodynamic coefficients (612 samples, $R^2$ 0.998, MAPE 2.9 percent).

## EDUCATION AND CERTIFICATIONS

**MSc Aerospace Engineering**, University of Salford, United Kingdom, 2023 (2:1). Dissertation: structural analysis of an aircraft engine bracket by the finite element method.

**BTech Aerospace Engineering**, Karunya Institute of Technology and Sciences, Coimbatore, India, 2021 (2:1).

**Certifications:** Master Diploma in Product Design (CADD Centre, 2022); Airbus Supply Chain Specialist (Airbus Beyond, 2026); Machine Learning Specialization (DeepLearning.AI and Stanford Online, 2026).